



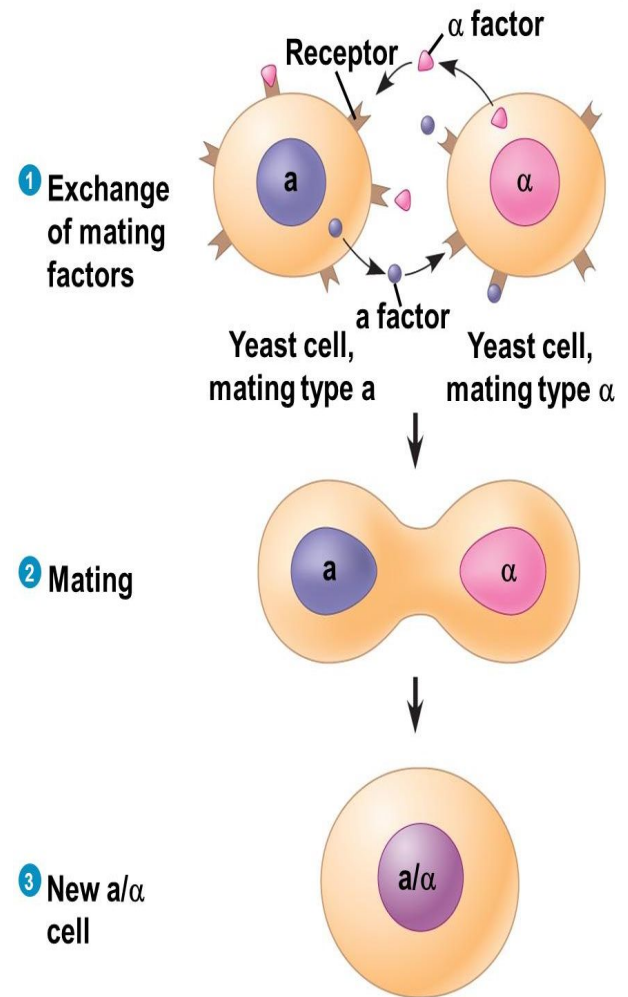
Introduction to Bioengineering Theory (BBL1020; 3-0-0)

Lecture-12: 1 Class

*Please listen Carefully
&
Keep your phone(s) on silent mode*

Overview: The Cellular Internet

- Cell-to-cell communication is essential for multicellular organisms
- Biologists have discovered some universal mechanisms of cellular regulation
- The combined effects of multiple signals determine cell response
- For example, multiple molecules crosstalk during vascular injury

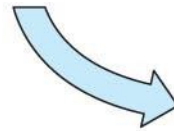
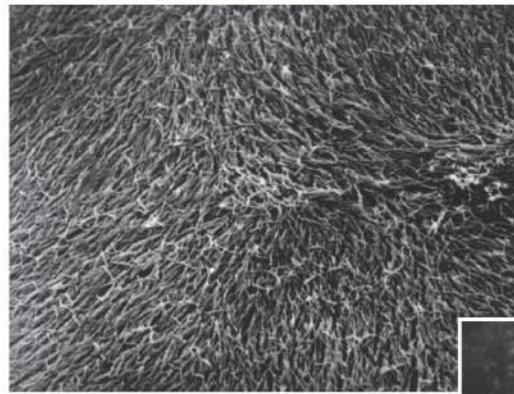


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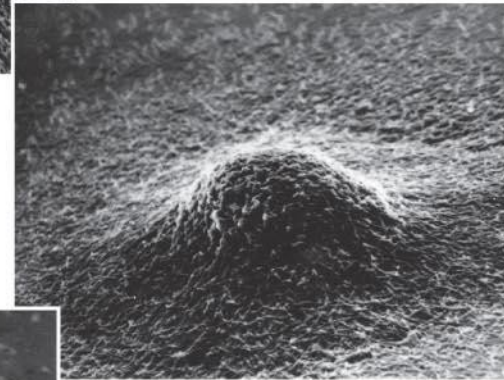
External signals are converted to responses within the cell

Communication among bacteria

1 Individual rod-shaped cells

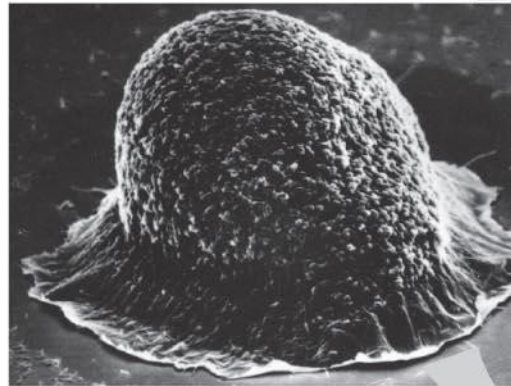


2 Aggregation in process

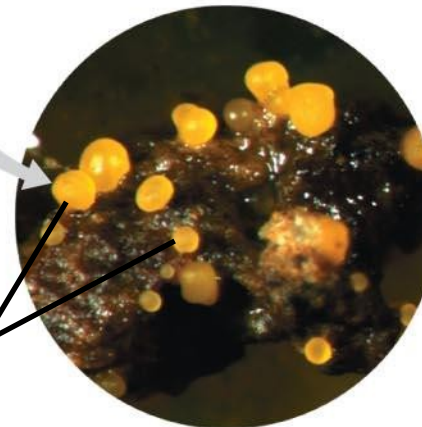


0.5 mm

3 Spore-forming structure (fruiting body)



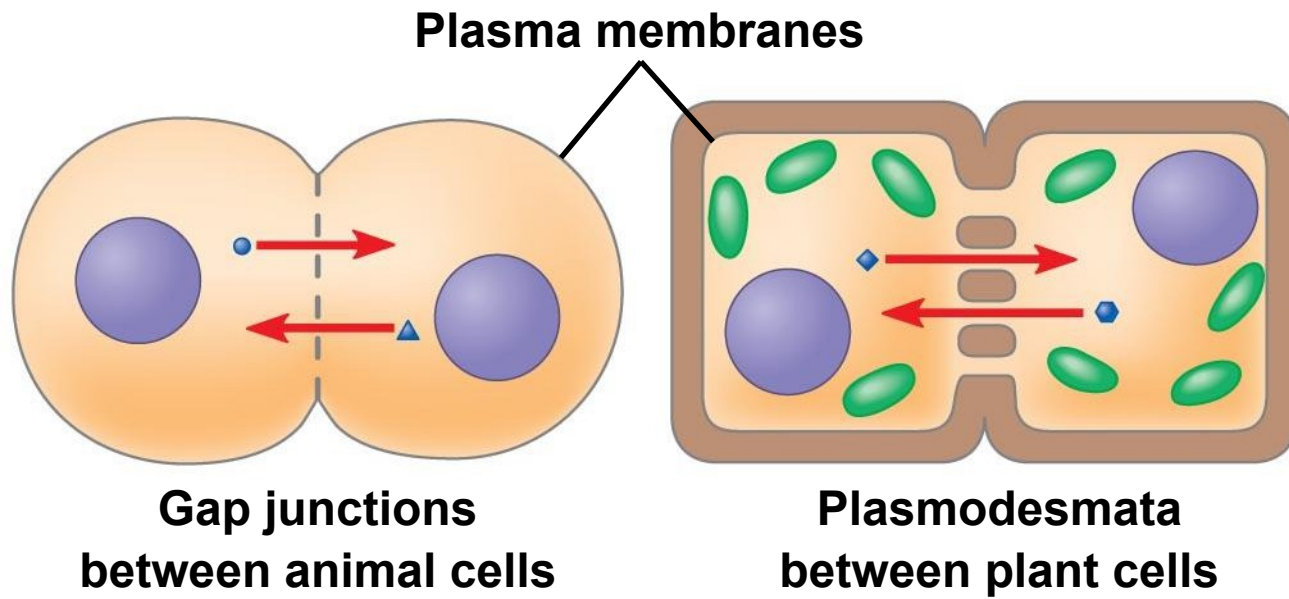
Fruiting bodies



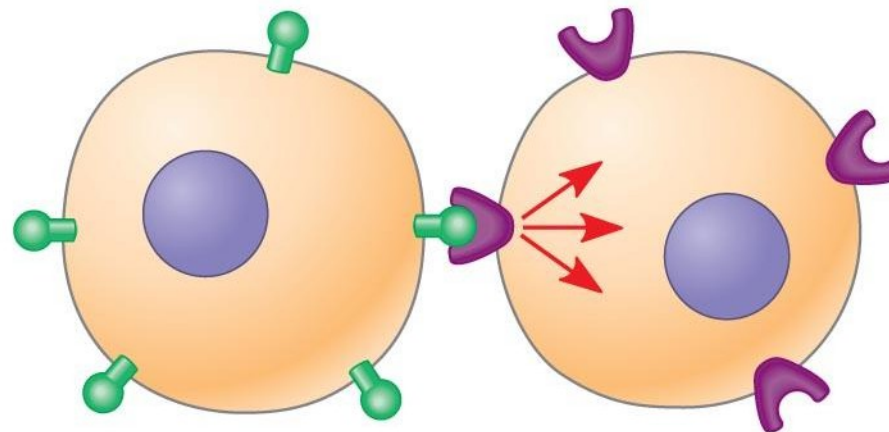
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- Pathway similarities suggest that ancestral signaling molecules evolved in prokaryotes and were modified later in eukaryotes
 - The concentration of signaling molecules allows bacteria to detect population density

Local and Long-Distance Signaling

- Cells in a multicellular organism communicate by chemical messengers
- Animal and plant cells have **cell junctions** that **directly connect** the **cytoplasm of adjacent cells**
- In local signaling, **animal cells** may communicate by **direct contact**, or cell-cell recognition



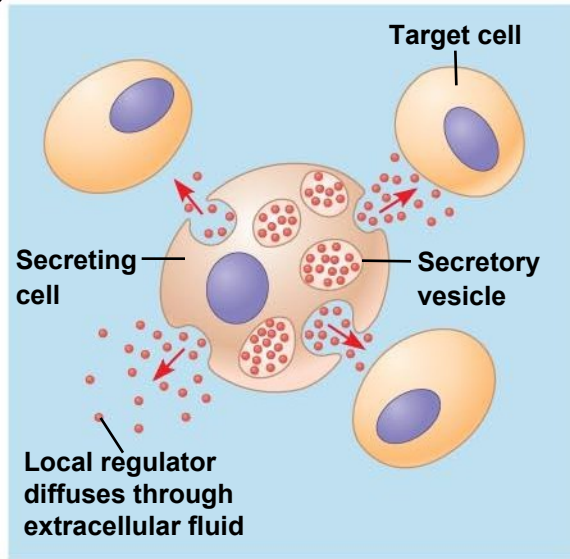
(a) Cell junctions



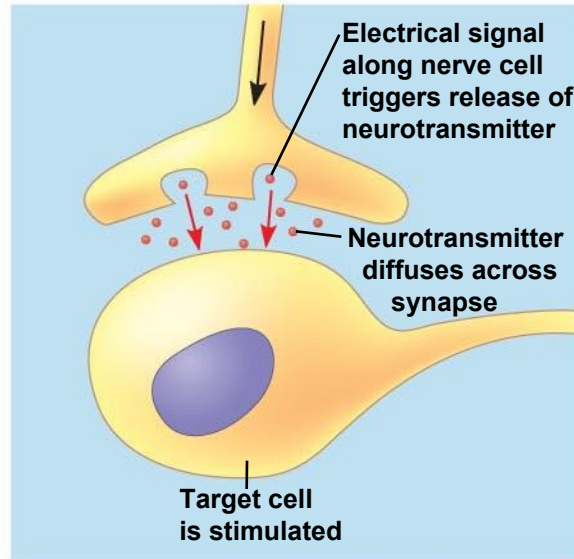
(b) Cell-cell recognition

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- In many other cases, animal cells communicate using **local regulators**, **messenger molecules** that **travel only short distances**
 - In **long-distance signaling**, plants and animals use chemicals called **hormones**

Local signaling

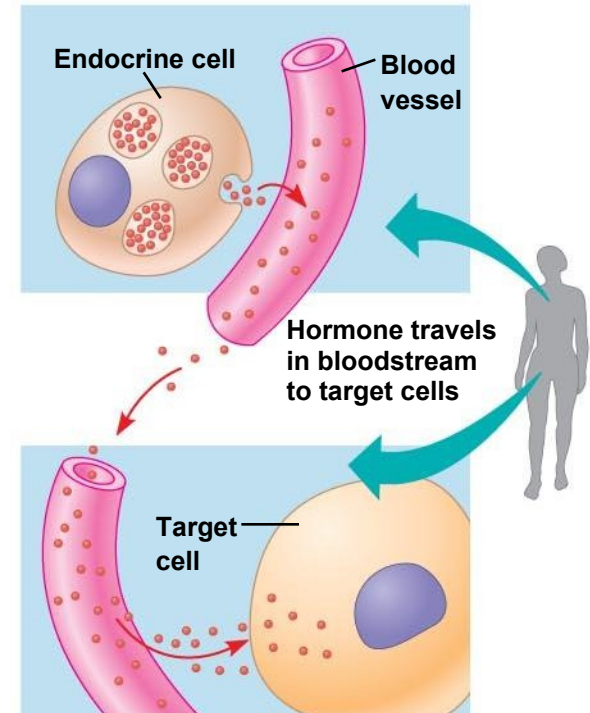


(a) Paracrine signaling



(b) Synaptic signaling

Long-distance signaling



(c) Hormonal signaling

The Three Stages of Cell Signaling: *A Preview*

- Earl W. Sutherland discovered how the hormone epinephrine acts on cells
- Sutherland suggested that cells receiving signals went through three processes:
 - **Reception**
 - **Transduction**
 - **Response**

Fig. 11-6-1

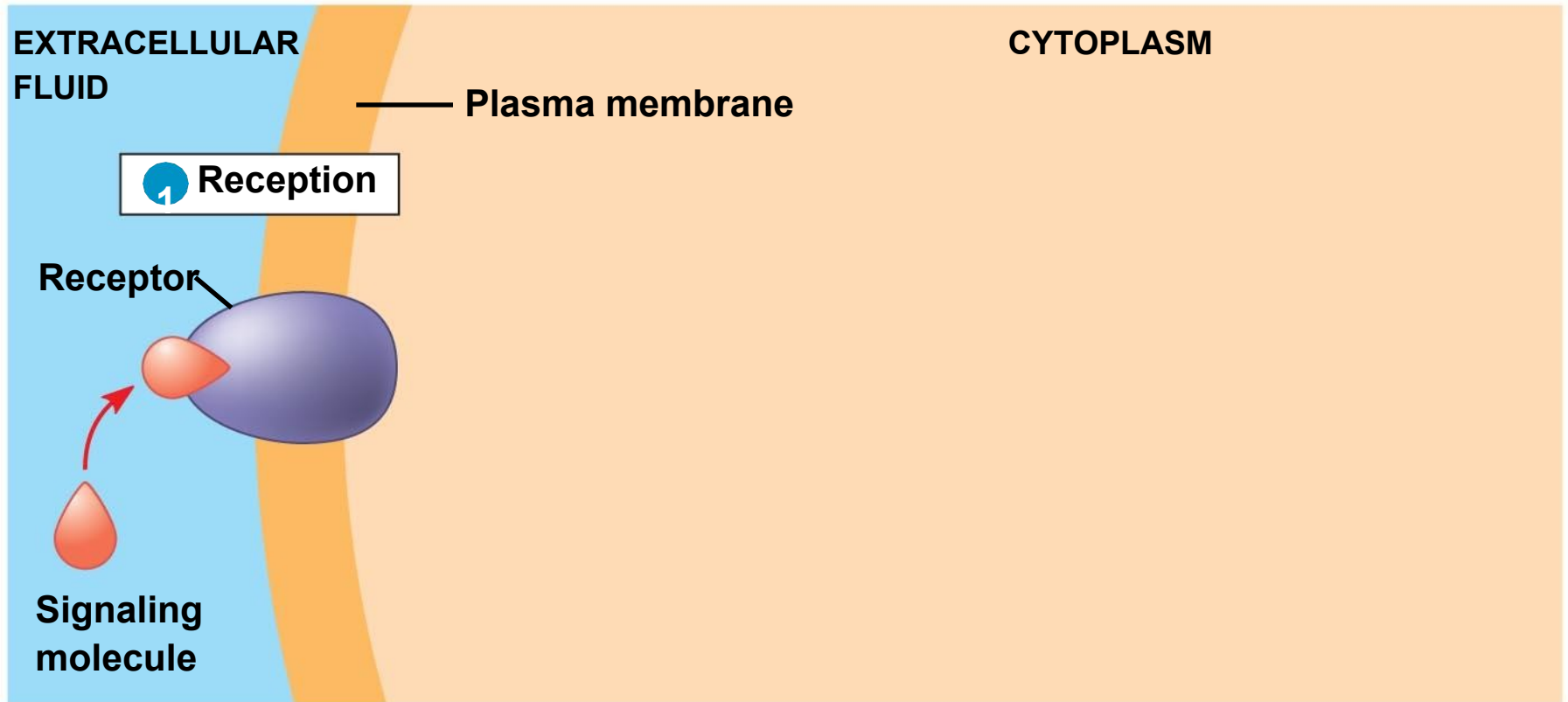


Fig. 11-6-2

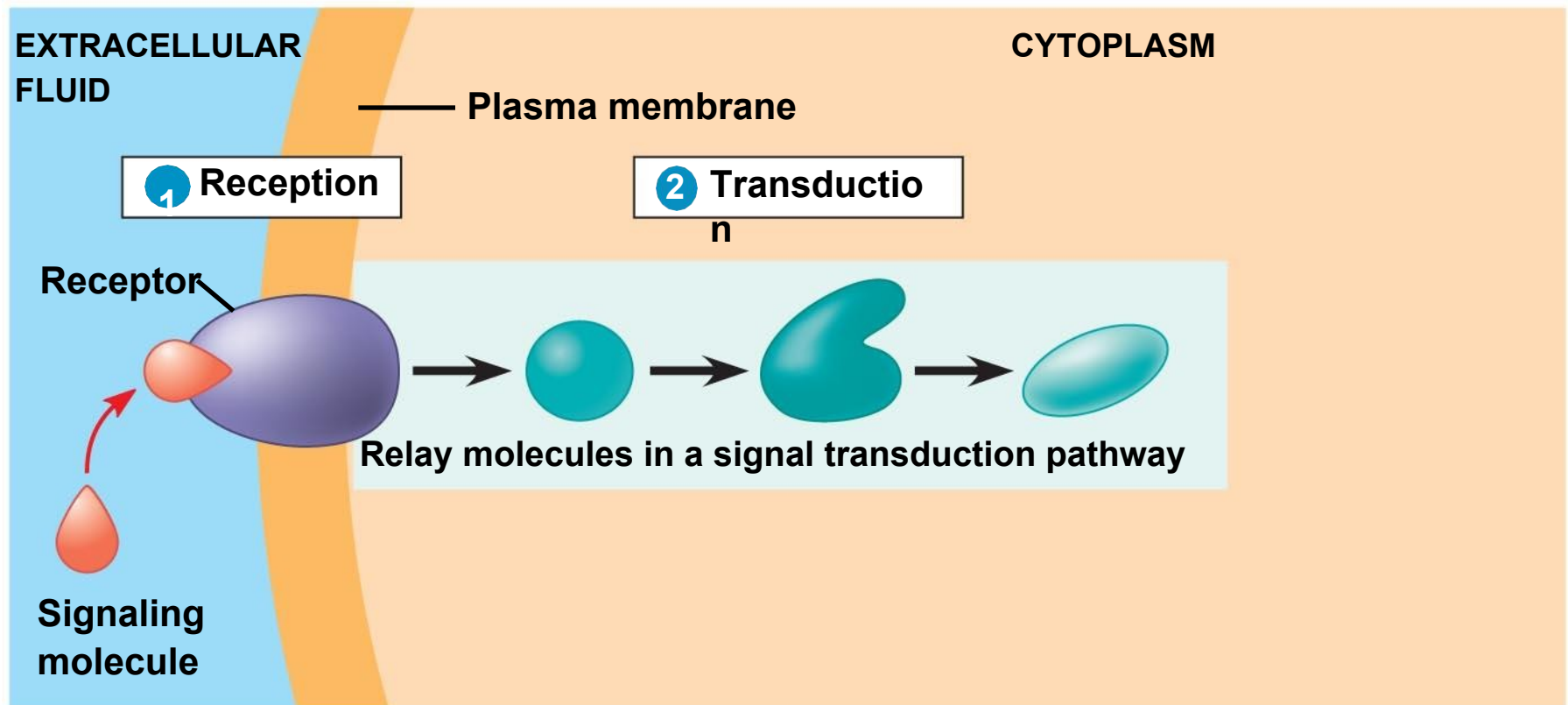
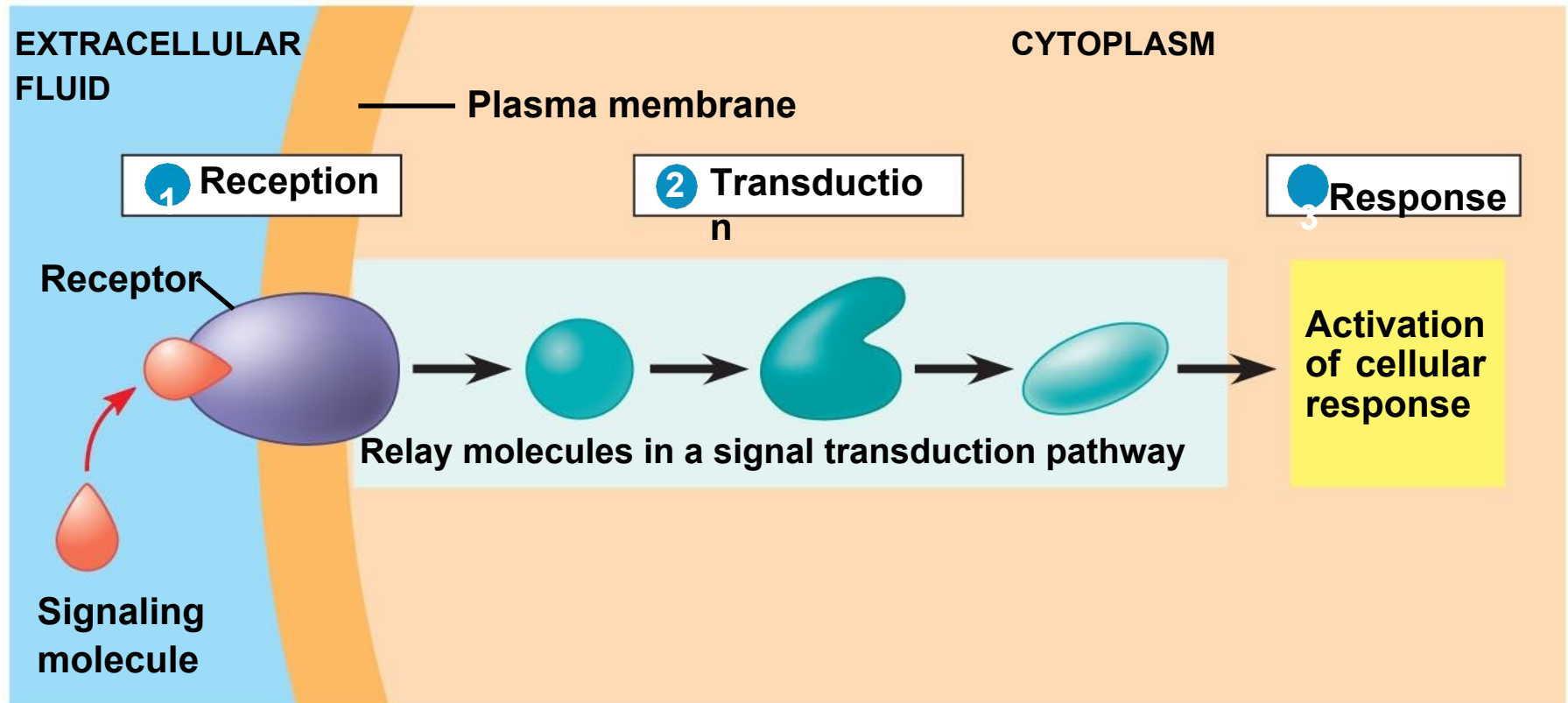


Fig. 11-6-3



Concept 11.2: Reception: A signal molecule binds to a receptor protein, causing it to change shape

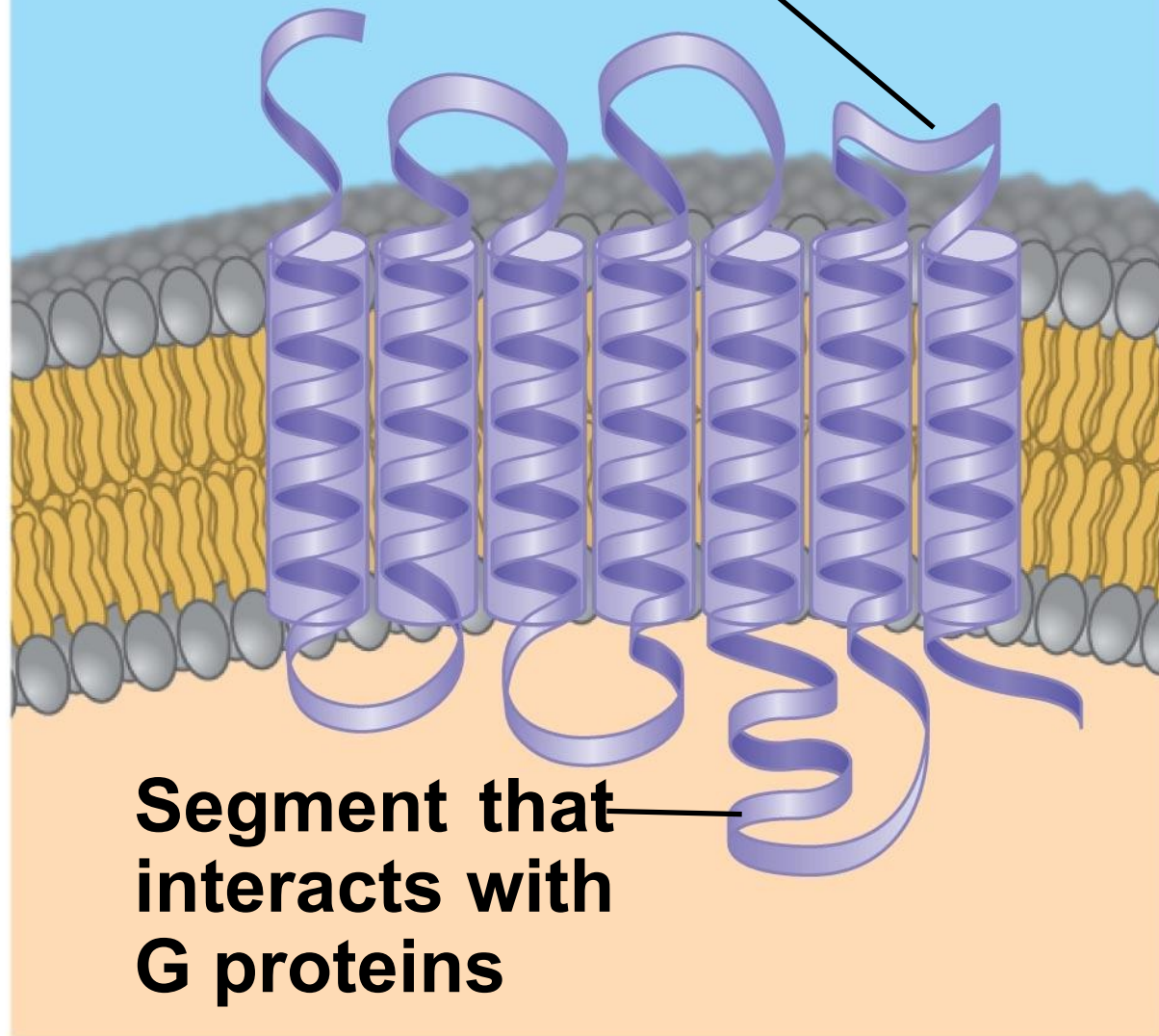
- The binding between a signal molecule (ligand) and receptor is highly specific
- A shape change in a receptor is often the initial transduction of the signal
- Most signal receptors are plasma membrane proteins

Receptors in the Plasma Membrane

- Most water-soluble signal molecules bind to specific sites on receptor proteins in the plasma membrane
- There are three main types of membrane receptors:
 - G protein-coupled receptors
 - Receptor tyrosine kinases
 - Ion channel receptors

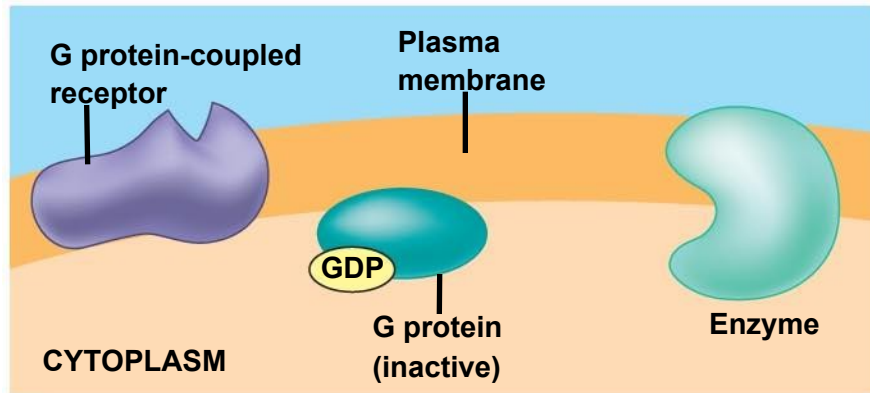
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- A **G protein-coupled receptor** is a plasma membrane receptor that works with the help of a **G protein**
 - The G protein acts as an **on/off switch**: **If GDP is bound to the G protein, the G protein is inactive**

Signaling-molecule binding site

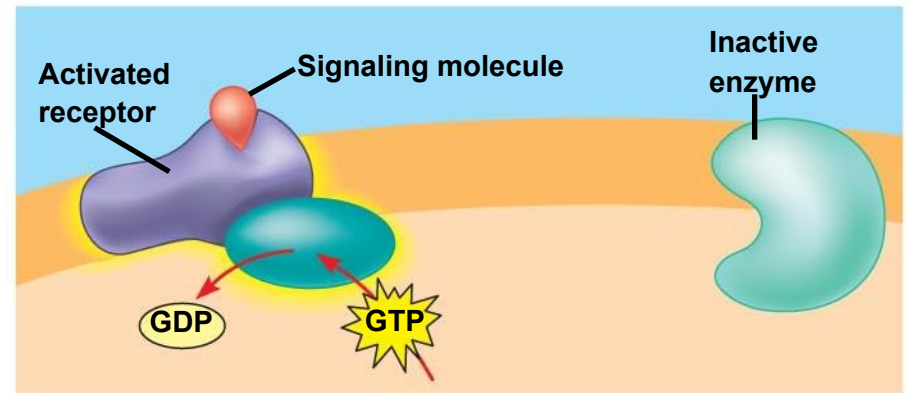


G protein-coupled receptor

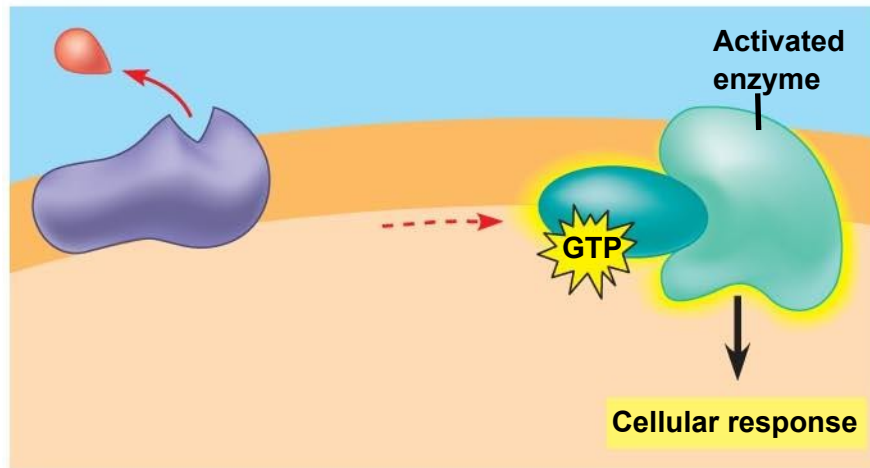
Fig. 11-7b



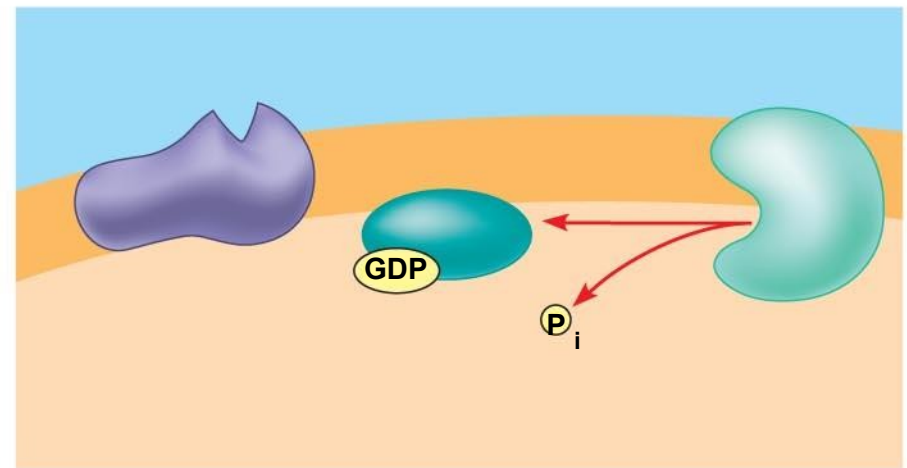
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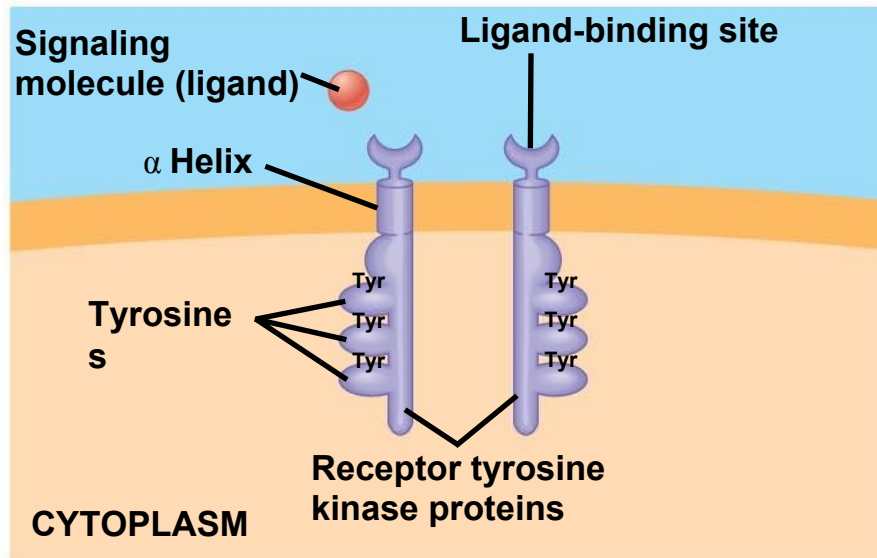
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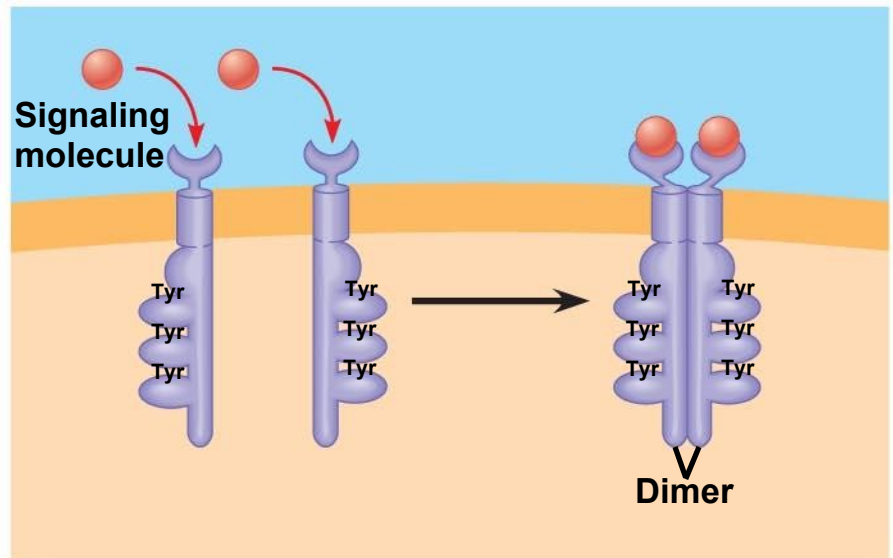
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- **Receptor tyrosine kinases** are membrane receptors that attach phosphates to tyrosines
 - A receptor tyrosine kinase can trigger **multiple signal transduction** pathways at once

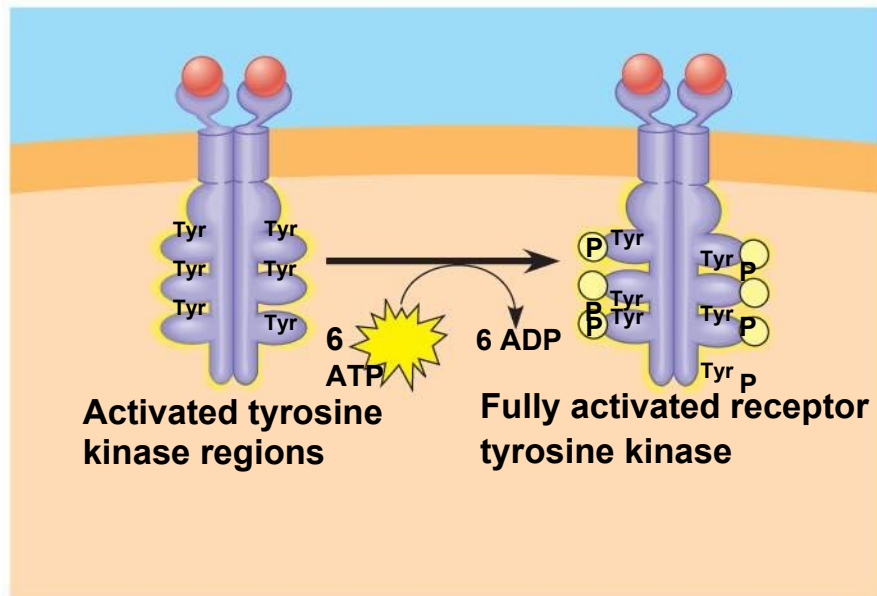
Fig. 11-7c



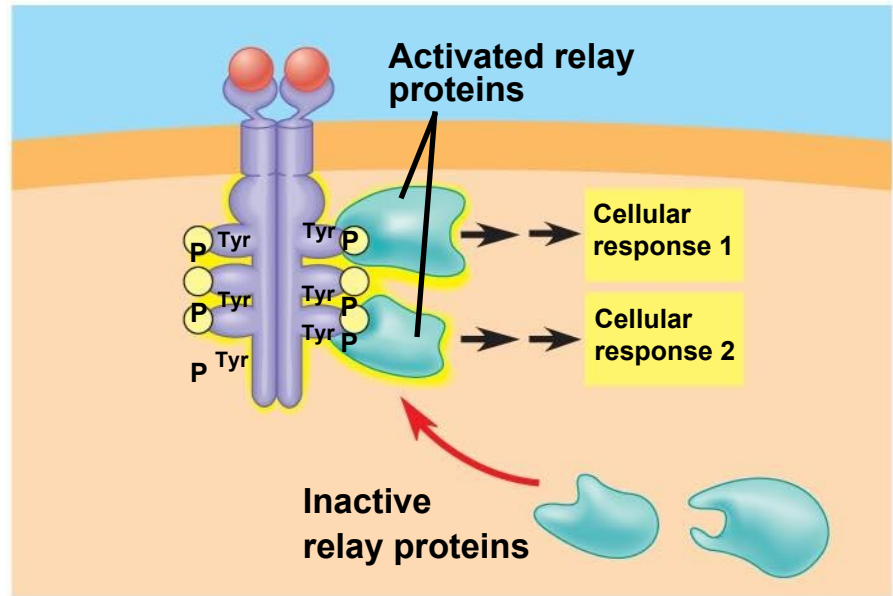
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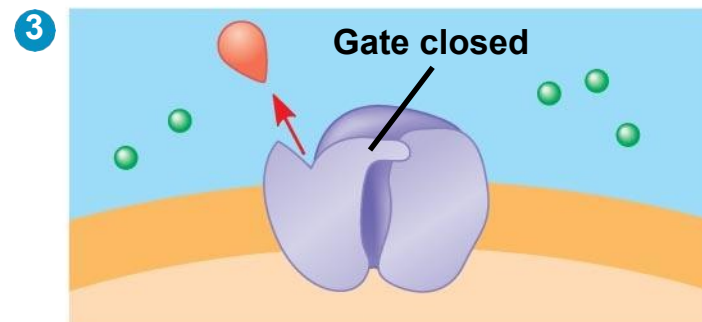
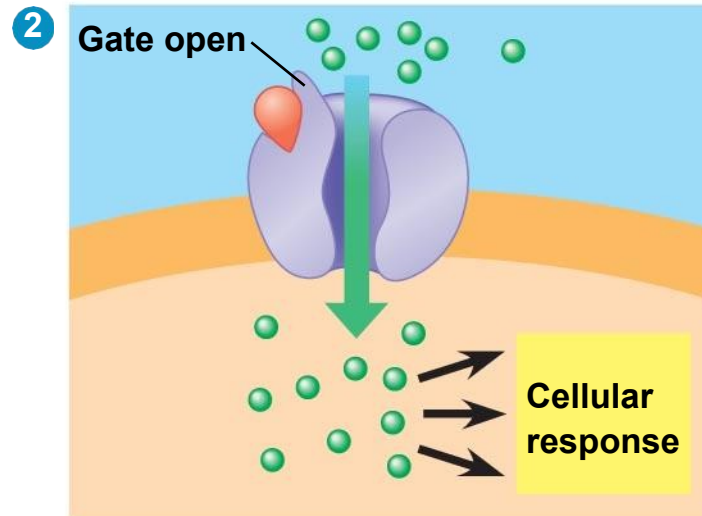
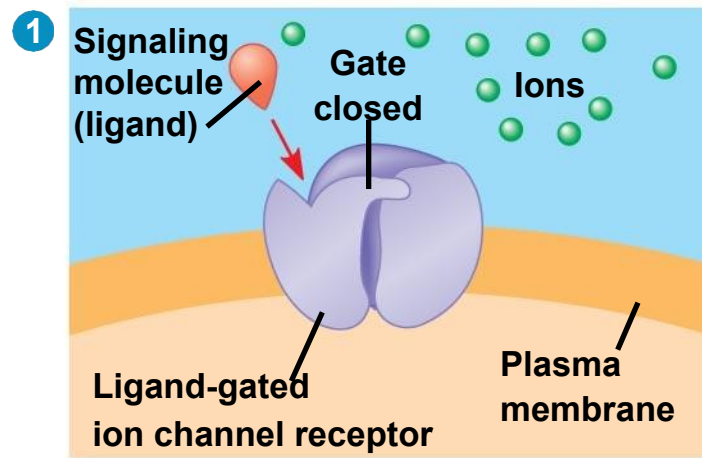
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- A **ligand-gated ion channel** receptor acts as a gate when the receptor changes shape
 - When a signal molecule binds as a ligand to the receptor, the gate allows specific ions, such as Na^+ or Ca^{2+} , through a channel in the receptor

Fig. 11-7d



Intracellular Receptors

- Some receptor proteins are intracellular, found in the **cytosol or nucleus of target cells**
- **Small or hydrophobic chemical messengers** can **readily cross the membrane and activate receptors**
- Examples of hydrophobic messengers are the steroid and thyroid hormones of animals
- An activated hormone-receptor complex can act as a transcription factor, turning on specific genes

